VIBHATH T K

Information Technology • Government Engineering College, Idukki

AI-augmented developer engineering autonomous desktop assistants, computer vision pipelines, and full-stack web applications with verified production prototypes.

Idukki, Kerala, India • vibhathcross@gmail.com • Portfolio • github.com/Vibhathcross • linkedin.com/in/vibhath-t-k

TECHNICAL SKILLS

Languages:	Python, JavaScript (ES6+), C, SQL, HTML5, CSS3
AI & Machine Learning:	LLM APIs (Gemini, Claude, Ollama), MediaPipe, OpenCV, ChromaDB, Resemblyzer, Q-Learning
Databases & Tools:	PostgreSQL (Supabase), SQLite, MongoDB, Git, Linux, Windows, FFmpeg, Postman

PROJECTS

ATLAS 2.0 — Advanced Autonomous Desktop AI Assistant & Automation Engine GitHub

Python, FastAPI, WebSockets, Three.js, PyQt6, MediaPipe, Resemblyzer, SQLite

- Architected an advanced autonomous desktop assistant evolved from Atlas OS, integrating FastAPI WebSockets, Three.js 3D visualization, MediaPipe 30 FPS air-mouse control, and Resemblyzer d-vector speaker biometrics to restrict voice commands to the authorized user.

ATLAS (v1) — Desktop Voice & OS Automation Assistant GitHub — Python, PyQt6, SpeechRecognition, Pyttsx3, WinAPI

- Engineered the foundational ATLAS desktop voice assistant with Python and PyQt6 to execute hands-free OS commands and window automations, serving as the direct functional prototype that motivated ATLAS 2.0.

Jasmin AI — Experimental AI Assistant (Trial Prototype) GitHub — Python, Flask, ChromaDB, Vosk, Whisper, WinAPI

- Developed Jasmin AI as an initial experimental trial exploring desktop AI assistance, integrating offline Vosk speech recognition, a Naive Bayes command classifier, and ChromaDB vector retrieval for local contextual recall.

Aether Carbon — Carbon Footprint Tracker GitHub | Live Demo | Prompt Wars Post — React 19, Vite, Tailwind CSS, Supabase, LLM APIs

- Developed an AI-augmented environmental tracking platform for **Google Prompt Wars** using React 19, Supabase RLS, and zero-shot LLM prompts to extract structured carbon metrics and mitigation advice from natural language activity journals.

CodSoft Projects (CODSOFT_TASKSNO) GitHub Repository | 3D Portfolio | Calculator | Landing Page

HTML5, Three.js, React 19, Supabase, Vite

- Built and deployed three production-grade web applications for the CodSoft internship, including a Three.js 3D WebGL portfolio, a responsive landing page, and a React 19 scientific calculator with Supabase history logging.

VidZone Video Editor — Client-Side Video Processing Tool GitHub — React 19, TypeScript, Electron, Vite, FFmpeg.wasm

- Engineered a zero-server desktop video editor using React 19, TypeScript, Electron, and FFmpeg.wasm that encodes lossless 1080p H.264 video natively on the client device.

Q-Learning Grid World Simulator GitHub — Python, Flask, NumPy, JavaScript, Chart.js

- Implemented a model-free Tabular Q-Learning simulation for a 5×5 discrete grid world with Bellman optimality updates and an interactive Flask web UI featuring live Chart.js training convergence curves.

TubeQueue — YouTube Upload & Cadence Scheduler GitHub — Python 3, Google API Client, OAuth 2.0, SQLite

- Created an automated YouTube video publishing daemon in Python utilizing Google OAuth 2.0 and Google's 5 MB chunked resumable upload protocol to ensure reliable scheduled releases across network interruptions.

Flagship Personal Portfolio GitHub | Live Site — Next.js, React 19, Tailwind CSS, WebGL Canvas

- Designed a physics-backed Next.js and React 19 engineering portfolio featuring GPU-accelerated canvas particle mechanics, interactive 3D card decks, and glassmorphic UI design.

AeroStream — Desktop Media Player GitHub — Electron, React 19, Vite, yt-dlp, FFmpeg

- Developed a desktop media player in Electron, React 19, and Vite that uses yt-dlp to deliver ad-free streaming, audio conversion, and local offline vault caching.

Padakkalam — Cultural Festival Web Portal GitHub | Live Site — HTML5, CSS3, JavaScript, JSON

- Built a cultural portal rendering dynamic JSON event schedules and game itineraries for Kerala Onam celebrations.

Thripura Business Webpage GitHub | Live Site — HTML5, CSS3, JavaScript

- Created a commercial digital showcase and service catalog portal for Thripura Press commercial printing operations.

EXPERIENCE

CodSoft — Web Development Intern — Virtual Internship

- Completed structured web development internship delivering full-stack and frontend modules with version control and automated deployments.

CERTIFICATIONS & VERIFICATION PROOFS

Cyber First Responder Awareness Programme — LinkedIn Proof
Kerala Police Cyber Division

Introduction to Generative AI — LinkedIn Proof
Google Cloud Training (Coursera)

Digital 101 Program (AI & Future Skills) — LinkedIn Proof
NASSCOM & MeitY (FutureSkills Prime)

30-Day Data Analytics MasterClass — LinkedIn Proof
NoviTech R&D

Web Development Internship Certificate — Certificate Proof
CodSoft

Foundations of Cybersecurity — LinkedIn Proof
Google (Coursera)

5-Day Web Development Bootcamp — LinkedIn Proof
µLearn IDK & INOVUS LABS IEDC

EDUCATION

Government Engineering College, Idukki — APJ Abdul Kalam Technological University (KTU) 2024 – 2028 (Expected)

Bachelor of Technology (B.Tech) in Information Technology

Coursework: Data Structures, Operating Systems, Database Management Systems, Computer Networks, Machine Learning Lab, Object-Oriented Programming (C/Python).

ACHIEVEMENTS

- **CLICK Cyber Resilience Initiative:** Delegate at the State-Level CLICK Inauguration (Kochi) organized by Kerala Police Cyber Division LinkedIn Post
- **Google Prompt Wars:** Participated with *Aether Carbon*, an AI-augmented environmental tracking tool LinkedIn Post
- **Open Source Developer:** Published and maintained 19 public repositories on GitHub across systems, AI, and web tools GitHub Repos